ARTHROPLASTY · NEET-SS / FET / INI-SS

Shoulder arthroplasty

1. Types of shoulder arthroplasty

Type	What is replaced	Key requirement	Main indications
Hemiarthroplasty	Humeral head only	Functioning cuff; glenoid acceptable	**Young patients** with AVN (no glenoid involvement), unreconstructable proximal humerus fractures in younger patients (now largely replaced by reverse in elderly), glenoid bone too poor for a component
Resurfacing (humeral head)	Humeral head surface	Adequate humeral bone stock	Young, active patients; AVN; preserves bone
Anatomic total shoulder arthroplasty (TSA)	Head + **polyethylene glenoid**	**Intact, functioning rotator cuff**	**Primary glenohumeral osteoarthritis** (best results), inflammatory arthritis with intact cuff

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Reverse total shoulder arthroplasty (RTSA)	**Glenosphere on glenoid, concave humeral cup**	**Functioning deltoid** and axillary nerve	**Cuff tear arthropathy**, massive irreparable cuff tears with pseudoparalysis, **proximal humerus fractures in the elderly (over about 70)**, failed arthroplasty, tumour, **B2/B3/C glenoids** with severe posterior erosion

2. Primary glenohumeral osteoarthritis

- **Pattern: posterior glenoid wear**, **posterior humeral head subluxation**, **anterior capsule and subscapularis contracture** (loss of **external rotation**: the commonest motion lost), **inferior humeral osteophyte ("goat's beard")**.
- **Walch classification of glenoid morphology (on axial CT):**
  - **A1** central minor erosion; **A2** central major erosion (medialisation),
  - **B1** posterior joint space narrowing, subchondral sclerosis, osteophytes; **B2 biconcave glenoid** (posterior erosion, posterior subluxation); **B3** monoconcave, posterior bone loss, **retroversion over about 15 degrees** and/or subluxation over 70%,
  - **C dysplastic glenoid, retroversion over 25 degrees**, not due to erosion,
  - **D** anteversion or anterior subluxation.
- **TSA outperforms hemiarthroplasty** for OA (pain relief, function, lower revision for glenoid erosion/pain). Correct moderate retroversion with **eccentric (high-side) reaming** (up to about 10-15

degrees), **augmented glenoid components**, bone grafting; severe B2/B3: consider **RTSA** (especially elderly).

- **Glenoid component: cemented all-polyethylene, pegged** components have lower radiolucency than keeled ones. **Glenoid loosening** is the **commonest long-term failure** of anatomic TSA (**rocking-horse phenomenon**: eccentric loading of the glenoid by a superiorly migrating head with cuff dysfunction).

3. Reverse total shoulder arthroplasty

Biomechanics (Grammont principles)

- **Medialises the centre of rotation** (onto the glenoid surface): increases the **deltoid moment arm** (more deltoid fibres recruited for abduction/elevation) and reduces shear on the glenoid component.
- **Lowers (distalises) the humerus**: **tensions the deltoid** (better lever).
- **Semi-constrained, fixed fulcrum**: allows elevation **without a functioning rotator cuff**.
- **Downsides of the classic Grammont design (155 degree neck-shaft angle, medialised): scapular notching**, limited rotation, loss of shoulder contour. **Modern designs lateralise** (glenoid-side via bony increased-offset RSA/BIO-RSA or metal augments; or humeral-side with a **135-145 degree** neck-shaft angle) to reduce notching, improve rotation and cuff (teres minor) tension.

Requirements

- **Functioning deltoid** and **intact axillary nerve** (the single absolute requirement), adequate glenoid bone stock.
- **External rotation:** depends on the **teres minor** (and infraspinatus). **Absent teres minor (positive hornblower's sign, external rotation lag)**: consider **latissimus dorsi (± teres major) transfer** with RTSA (L'Episcopo-type).

Complications of RTSA

- **Scapular notching** (commonest radiological finding; **Sirveaux classification grades 1-4**: erosion of the inferior scapular neck by the humeral cup in adduction; reduced by **inferior glenosphere placement, inferior tilt, lateralisation, larger glenosphere, lower neck-shaft angle**).
- **Instability/dislocation** (commonest complication in some series; **subscapularis deficiency**, inadequate soft-tissue tension, **medialised designs**, deltopectoral approach).
- **Acromial and scapular spine stress fractures** (**Levy classification**: type 1 acromion, type 2 at the base of the acromion/scapular spine origin of the deltoid, type 3 scapular spine): sudden pain and loss of elevation after a period of good function; associated with **over-lengthening (over-tensioning the deltoid), lateralisation**, osteoporosis.
- **Infection** (higher than anatomic TSA): ***Cutibacterium (Propionibacterium) acnes*** (**commonest in shoulder PJI**; **male sex**, young, hairy skin, sebaceous glands; slow-growing: **hold cultures for 14 days**; **benzoyl peroxide** skin preparation reduces load).
- **Glenoid baseplate loosening**, **nerve injury** (**axillary nerve**; brachial plexus traction with lengthening), **haematoma**, **humeral fracture**.

4. Proximal humerus fractures and arthroplasty

- **Hemiarthroplasty** results depend on **tuberosity healing** in the correct position (tuberosity malunion/non-union causes poor elevation): technically demanding; unpredictable function.
- **RTSA** gives **more reliable forward elevation and lower revision** than hemiarthroplasty in **elderly patients (over about 70) with 3/4-part fractures, head-split, fracture-dislocations**; **repair tuberosities** anyway (improves external rotation and active elevation). **PROFHER trial:** no advantage of surgery overall over non-

operative management for most displaced proximal humerus fractures (in elderly with surgical neck involvement).

- **Hertel (LEGO) criteria** for **head ischaemia** after fracture: **metaphyseal head extension under 8 mm**, **disrupted medial hinge (over 2 mm displacement)**, **anatomical neck fracture pattern**: high predictive value when combined.

5. Cuff tear arthropathy (CTA)

- Massive chronic cuff tear: superior migration of the humeral head, **acetabularisation of the acromion** (concave undersurface), **femoralisation of the humeral head** (rounding of the tuberosities), loss of the acromiohumeral interval (**under 7 mm**: suggests massive cuff tear), glenohumeral arthritis, **Milwaukee shoulder** (hydroxyapatite crystal-associated, elderly women, large blood-stained effusions).
- **Hamada classification: grade 1** AHI over 6 mm; **2** AHI 5 mm or less; **3** acetabularisation; **4** glenohumeral arthritis (4A without, 4B with acetabularisation); **5** humeral head collapse.
- **Seebauer classification** (biomechanics: centred vs decentred, stable vs unstable): **IA, IB** (centred), **IIA, IIB** (decentred: superior migration / anterosuperior escape: IIB = **anterosuperior escape** with **coracoacromial arch** loss: **avoid acromioplasty/CA ligament release** in massive tears!).
- **RTSA** is the treatment of choice for symptomatic CTA with pseudoparalysis; **hemiarthroplasty with a CTA head** (extended articular surface) in younger/low-demand patients with preserved elevation and intact coracoacromial arch.

6. Other indications and specifics

- **Rheumatoid arthritis:** central glenoid erosion (medialisation), cuff tears common; **RTSA** if cuff deficient; check the cervical spine.

- **AVN of the humeral head (Cruess classification, stages I-V, like Ficat):** steroids, sickle cell, alcohol, post-trauma; **hemiarthroplasty/resurfacing** if the glenoid is intact (stage III-IV), **TSA** if glenoid involved (stage V).
- **Instability arthropathy** (after previous stabilisation, e.g. **capsulorrhaphy arthropathy** from over-tight anterior repair (Putti-Platt), **hardware arthropathy**).
- **Post-infective arthritis**, **post-traumatic arthritis** (fracture sequelae: **Boileau classification** types 1-4; **types 3 (non-union of the surgical neck) and 4 (severe tuberosity malunion)** do poorly with anatomic arthroplasty unless osteotomy: favour **RTSA**).
- **Approach: deltopectoral** (commonest; cephalic vein, **subscapularis management: tenotomy, peel or lesser tuberosity osteotomy**; subscapularis failure after anatomic TSA: anterior instability, weakness: **lift-off and belly-press signs**). **Superolateral (Neviaser-Mackenzie)** approach for some RTSA (less instability, worse glenoid exposure, risk to the axillary nerve with deltoid split over about 5 cm distal to the acromion).
- **Stemless** humeral components (metaphyseal fixation) in anatomic TSA: preserve bone.
- **Rehabilitation:** protect subscapularis repair (limit external rotation initially) in anatomic TSA.

7. Outcomes

- **Anatomic TSA for OA:** about **90%+ survival at 10 years**; best outcomes of any shoulder arthroplasty for pain relief.
- **RTSA:** good pain relief and elevation for CTA (**forward elevation typically to about 120-140 degrees**), **limited internal and external rotation**; survival about 90% at 10 years; worse outcomes in **revision RTSA** and post-traumatic sequelae.
- **Hemiarthroplasty for OA: progressive glenoid erosion** (painful), inferior to TSA.

Exam pearl: **Anatomic TSA needs an intact cuff; RTSA needs a working deltoid (axillary nerve). Commonest late failure of anatomic TSA = glenoid loosening (rocking horse). Commonest radiological finding after RTSA = scapular notching.**

Exam pearl: **Grammont: medialise and distalise the centre of rotation to increase the deltoid lever arm. Loss of teres minor (hornblower's sign): add latissimus dorsi transfer. Acromial stress fracture after RTSA: sudden pain and loss of elevation.**

Exam pearl: **Shoulder PJI: C. acnes; hold cultures 14 days. Walch B2 = biconcave glenoid. Elderly 3/4-part proximal humerus fracture: RTSA more reliable than hemiarthroplasty (tuberosity healing).**

Written by Claude for Ortho preSSaleel · updated 2026-11-14. Check doubtful points in a standard textbook.